

Maxwell–Teleparallel Algebra & Chern–Simons Theory

Reference Notes

Cohomology

de Rham Cohomology

Nilpotency $d^2 = 0$ yields the cochain complex $\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \dots$ with $\text{Im}(d_n) \subset \ker(d_{n+1})$. Define

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k \rightarrow \Omega^{k+1})}{\text{Im}(d : \Omega^{k-1} \rightarrow \Omega^k)}.$$

Closed forms not globally exact detect global topology. The de Rham theorem gives $H_{\text{dR}}^k(M) \cong H^k(M, \mathbb{R})$; Poincaré duality $H^k \cong H_{\text{dR}}^{n-k}$ for closed orientable M^n .

Čech Cohomology

Given open cover $\mathcal{U} = \{U_i\}$, a Čech p -cochain assigns values on $(p+1)$ -fold overlaps. The coboundary δ satisfies $\delta^2 = 0$; for good covers $\check{H}^p(M, \mathbb{R}) \cong H_{\text{dR}}^p(M)$. Transition functions $g_{ij} : U_i \cap U_j \rightarrow G$ satisfying the cocycle condition $g_{ij}g_{jk}g_{ki} = 1$ classify bundles and gauge sectors.

Chevalley–Eilenberg Cohomology

For $\mathfrak{g}$ -module V , set $C^p(\mathfrak{g}; V) = \text{Hom}(\Lambda^p \mathfrak{g}, V)$. The differential

$$(d_{\text{CE}}\omega)(x_1, \dots, x_{p+1}) = \sum_{i < j} (-1)^{i+j} \omega([x_i, x_j], \dots) + \rho\text{-terms}$$

satisfies $d_{\text{CE}}^2 = 0$ by the Jacobi identity. Key classes: $H^0 = \mathbb{R}$; $H^1 = (\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}])^*$ (abelianisation); the canonical three-cocycle

$$\omega_3(x, y, z) = \langle x, [y, z] \rangle \in H^3(\mathfrak{g}; \mathbb{R})$$

for simple $\mathfrak{g}$ (unique up to scale) is the origin of WZW and CS. The Maurer–Cartan equation $d\Theta + \frac{1}{2}[\Theta \wedge \Theta] = 0$ encodes H^2 .

Whitehead Lemmas. For $\mathfrak{g}$ semi-simple and V finite-dimensional over char 0: $H^1(\mathfrak{g}; V) = H^2(\mathfrak{g}; V) = 0$. Consequently: every derivation is inner ($H^1 = 0$), every central extension splits ($H^2 = 0$). The Poincaré algebra is *not* semi-simple (translation ideal is abelian), so non-trivial extensions — in particular the Maxwell algebra via a class in $H^2(\mathfrak{poinc}; \mathfrak{h})$ — are permitted. Quantisation of the CS/WZW level k follows from $H^3(G, \mathbb{Z})$ integrality.